UNIVERSITY OF LONDON

MSci/MSc EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS4100/CS5100: Data Analysis
CS4100R/CS5100R: Data Analysis
CS5100O: Data Analysis

— for FIRSTSIT/RESIT CANDIDATES—

Time Allowed: **TWO hours**

Please answer **ALL** questions

EPMS, CS, PH, MT, and EE calculators permitted

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1. **Decision Trees (25 marks)**

   (a) Explain why Decision Trees can be used for regression and binary or multi-class classification. [5 marks]

   (b) Focus on classification Decision Trees. What is a region's *purity*? How do you measure it using the Entropy or the Gini index? [5 marks]

   (c) Give the mathematical formula of the Entropy (or the Gini index) of $D = \{(X_n, Y_n)\}_{n=1}^{N}$, where $X_n \in \mathbb{R}$ and $Y_n \in \{\texttt{yes}, \texttt{no}\}$. **Hint**: Write the Entropy in terms of the class probabilities, $p_{\texttt{yes}} = p_{\texttt{yes}}(D)$ and $p_{\texttt{no}} = 1 - p_{\texttt{yes}}$, and specify how you obtain them from the samples in D. [5 marks]

   (d) Discuss the flexibility of Decision Trees in terms of the number of terminal nodes. Give an example of a technique you can use to mitigate overfitting problems. [5 marks]

   (e) Consider the following data set,

$$\mathcal{D} = \left\{ (1, \texttt{yes}), \left(\frac{1}{2}, \texttt{yes}\right), \left(\frac{1}{3}, \texttt{no}\right) \right\}$$

   Compute the binary partition that maximizes the purity of the obtained regions and use it to predict the probability of observing $Y = \texttt{yes}$ for an object with attribute $X = 0$. **Hint**: You only need to compare 2 thresholds, $\frac{1}{3} \leq \xi_1 < \frac{1}{2}$ and $\frac{1}{2} \leq \xi_2 < 1$. [5 marks]

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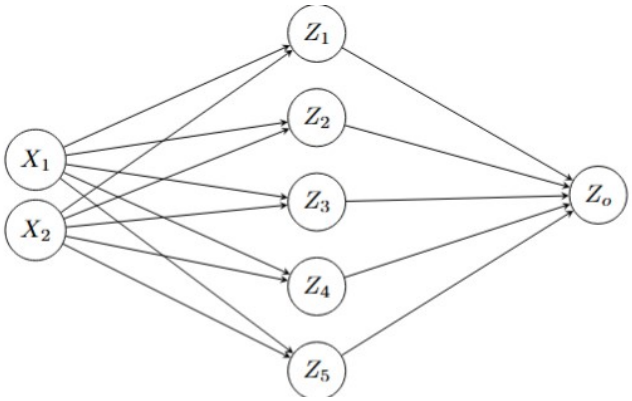


Figure 1: $f_5 : \mathbb{R}^2 \to \mathbb{R}_+$ is a Shallow Neural Network with 2 inputs, 5 hidden neurons, and 1 output.

2. **Neural Networks (25 marks)**

   (a) What is a Neural Network? Explain the difference between input, output, and hidden neurons. [5 marks]

   (b) The Shallow Neural Network represented in Figure 1, $f_5 : \mathbb{R}^2 \to \mathbb{R}_+$, has 21 free parameters. Would the number of parameters be higher if the network had 3 instead of 2 inputs? Justify your answer. [5 marks]

   (c) Briefly explain why f_5 can be expected to be more flexible than $f_{linear}(X) = \lambda_0 + \lambda_1 X_1 + \lambda_2 X_2$, a linear regression model with an analogous input-output structure. [5 marks]

   (d) Neural Networks are often trained by minimizing an objective function. Give the mathematical formula of an objective function that you can use for training f_5. **Hint:** The network output is a real number. [5 marks]

   (e) Assume the parameters of the hidden neurons are $\lambda_i = [i, 1, \frac{1}{2}]^T$, $i = 1, \ldots, 5$, the parameter of the output neuron is $\lambda_o = [-5, 4, 3, 2, 1, 0]$ and all neurons have a ReLU activation function. Evaluate the network output for input $X = [-1, -2]$. **Hint**: Start by evaluating the value of the hidden neurons, $Z_i = \lambda_{i0} + \lambda_{i1} X_1 + \lambda_{i2} X_2$, and then compute the output of the network,

$$\hat{Y} = \mathrm{ReLU}\left(\lambda_{o0} + \sum_{i=1}^{5} \lambda_{oi} \mathrm{ReLU}(Z_i)\right)$$

   where $\mathrm{ReLU}(t) = \max\{0, t\}$. [5 marks]

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3. **Ensemble Methods (16 marks)**

(a) Is it possible to overfit the bagging meta-algorithm by combining too many decision trees (or other base prediction rules)? Is it possible to overfit the boosting meta-algorithm by combining too many decision trees (or other base prediction rules)? Give a brief (one-sentence) justification of your answers. [5 marks]

(b) Consider the following data set with 4 data points in the form of (x, y):

$$\mathcal{D} = \{(0, 2),\ (2, 3),\ (4, 6),\ (5, 10)\}.$$

This table shows five bootstrapped training sets (BTS) and the predictions made by the model trained on each BTS:

BTS	Predictions at $x =$			
	0	2	4	5
$\{(0,2),(0,2),(2,3),(2,3)\}$	2	4	6	12
$\{(0,2),(2,3),(5,10),(5,10)\}$	2	3	5	9
$\{(2,3),(2,3),(2,3),(2,3)\}$	-1	3	4	6
$\{(4,6),(4,6),(4,6),(5,10)\}$	4	4	6	11
$\{(0,2),(0,2),(4,6),(5,10)\}$	2	3	5	9

i. What are the OOB (out-of-bag) predictions for $x = 2$ and $x = 5$? [6 marks]

ii. The OOB predictions for $x = 0$ and $x = 4$ are 1.5 and 5, respectively. Use this information and the results in part i. to calculate the OOB MSE. [5 marks]

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4. **Information Retrieval (34 marks)**

(a) Give a brief definition of information retrieval. [3 marks]

(b) Define the *term frequency* of a term in a document. [2 marks]

(c) Define the *document frequency* and *inverse document frequency* of a term in a collection. [5 marks]

(d) Give the definition of the tf-idf weighting scheme. What is the intuition behind this scheme? [7 marks]

(e) Explain how you obtain a vector representation of a document (known as document vector). [2 marks]

(f) Define the cosine similarity between two documents. What is the main advantage of the cosine similarity between documents compared with the Euclidean distance between them? [5 marks]

(g) Consider a collection consisting of 4 web pages, A, B, C, and D. These are the term frequencies in them:

term	A	B	C	D
learning	24	7	0	2
Python	4	12	0	11
R	6	0	1	2

The web pages in the collection contain only those terms. Rank the web pages according to their relevance to the query `learning Python`. When computing the document vectors use tf-idf, as usual. You may round all intermediate results to two decimal places. [10 marks]

END

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